NOVA SCHOOL OF SCIENCE & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE

MEI - Master in Computer Science and Engineering

Internet Applications Design and Implementation

Week 3 - Software architecture

Information

The dates of Test 1 and Test 2 were updated as instructed by the Master's program coordination.

Test 1	30 Apr
Test 2	23 May

Both the site (github) and the slides from week 1 were updated.

This is also in CLIP.

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Software architecture

“The software architecture of a system is the set of structures needed to reason about the system, which comprise software elements, relations among them, and properties of both.”

(in Software Architectures in Practice)

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Software architecture

Three kinds of structures:

- Module structures
  code and data units (implementation)
- Component-and-connector structures
  elements that have runtime behaviour (components) and interactions (connectors)
- Allocation structures
  mapping between software structures and system's non-software structures (deployment and/or organizational)

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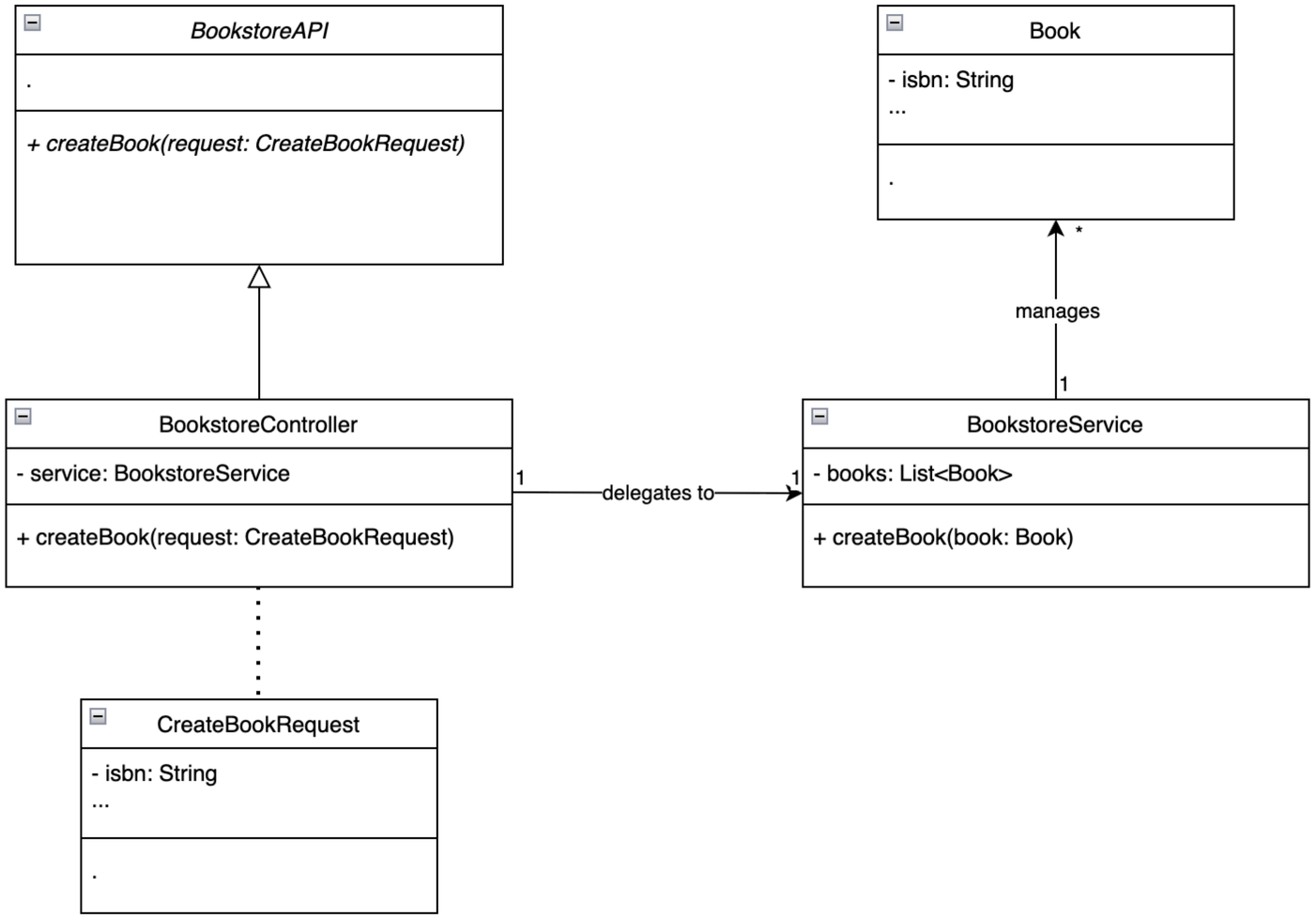
Software architecture

Implementation

Examples:

UML Class Diagram

E/R Diagram



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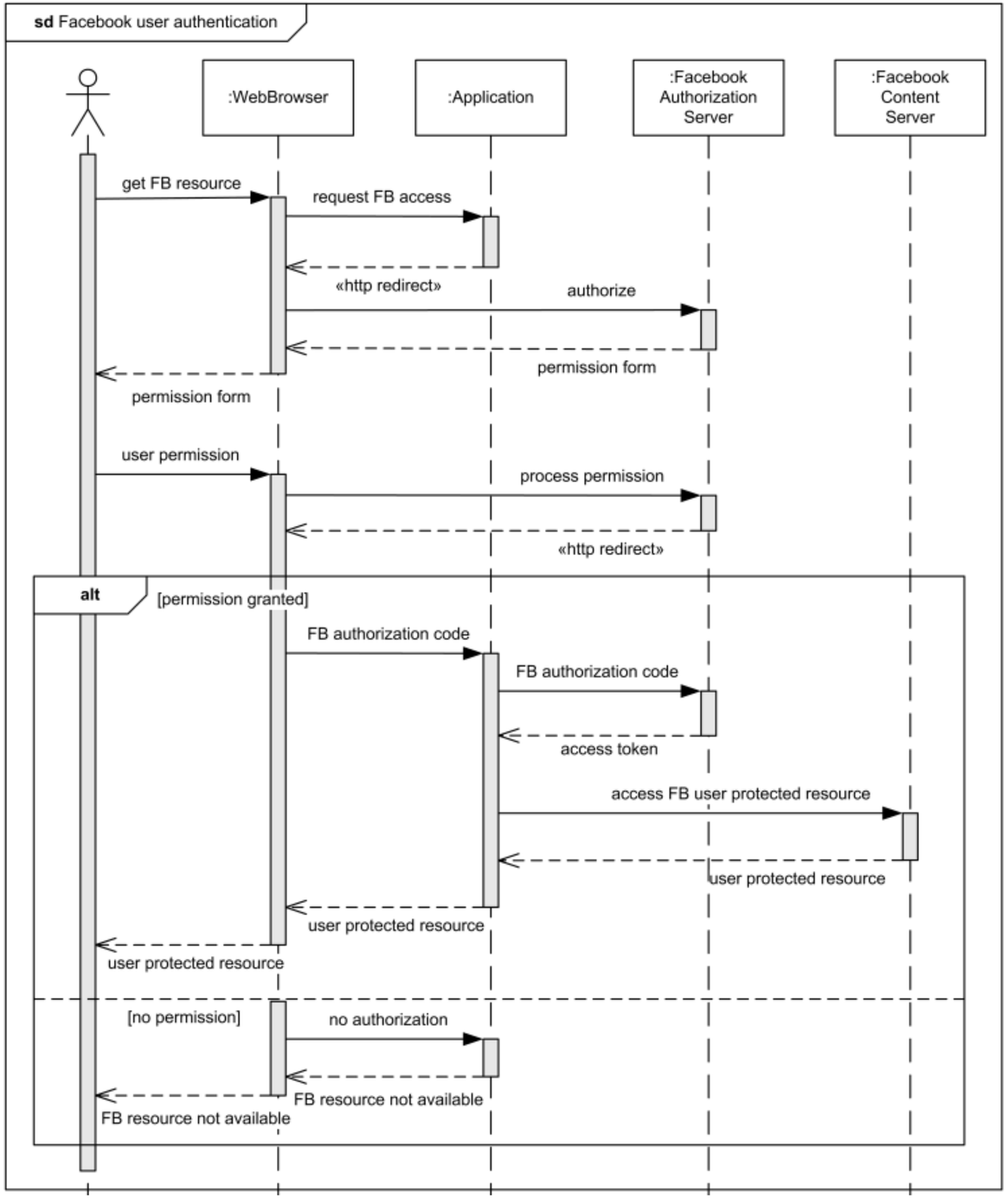

Software architecture

Runtime

Examples:

UML Sequence Diagram

UML Activity Diagram

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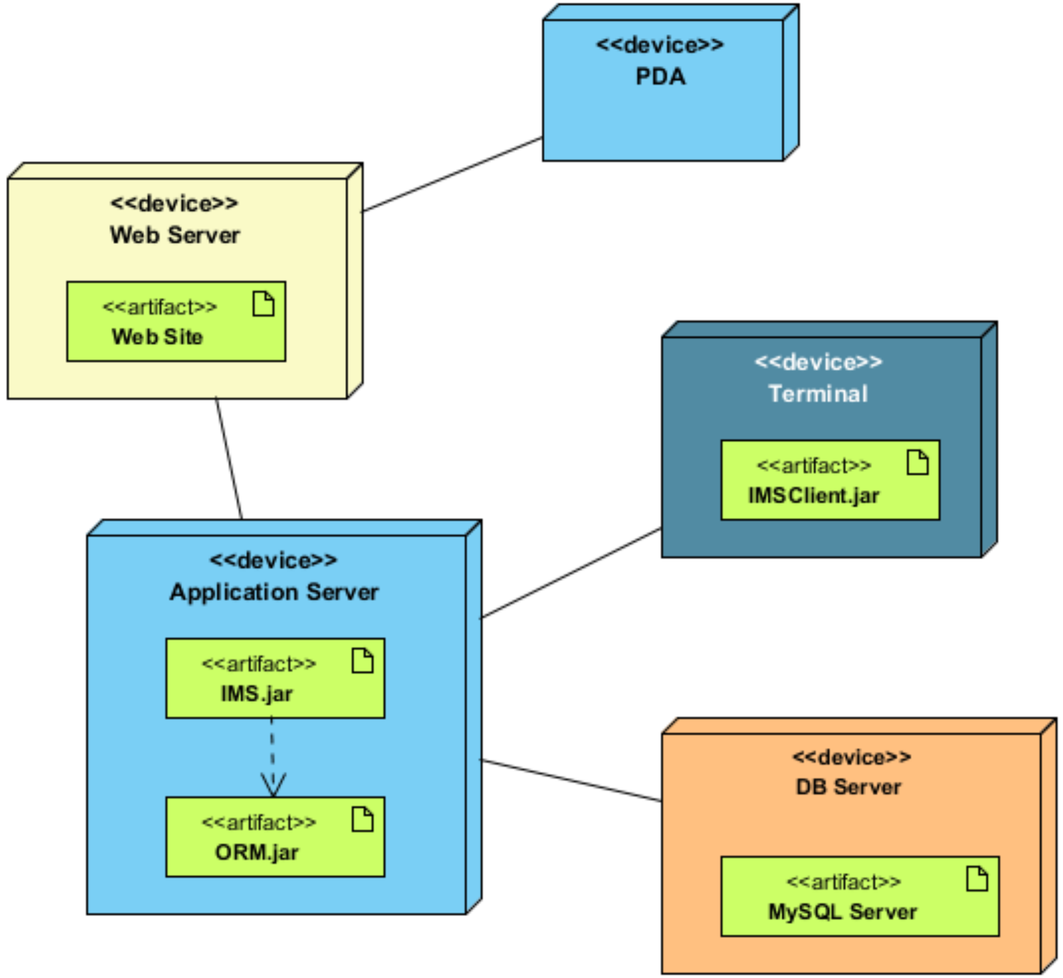
Deployment

Examples:

UML Deployment Diagram



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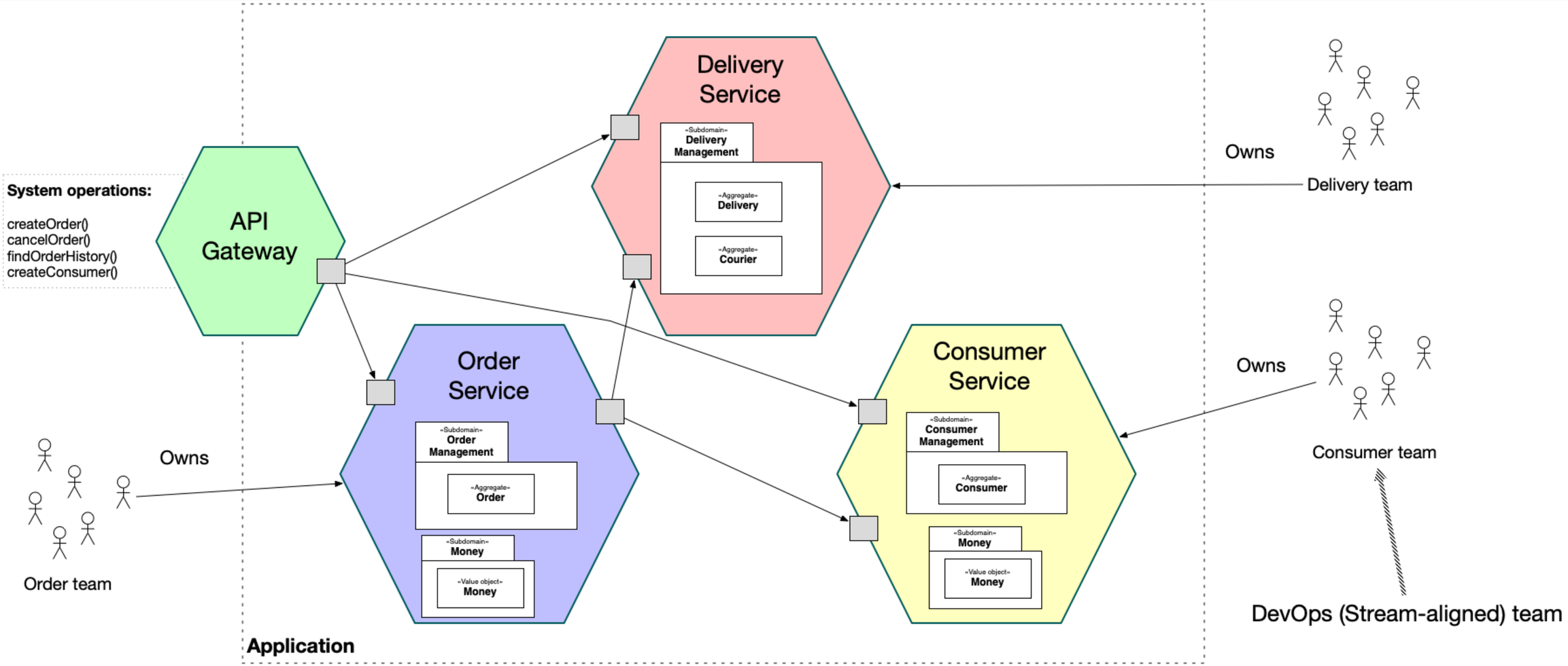
Software architecture

Organizational



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Software architecture

What is a good architecture?

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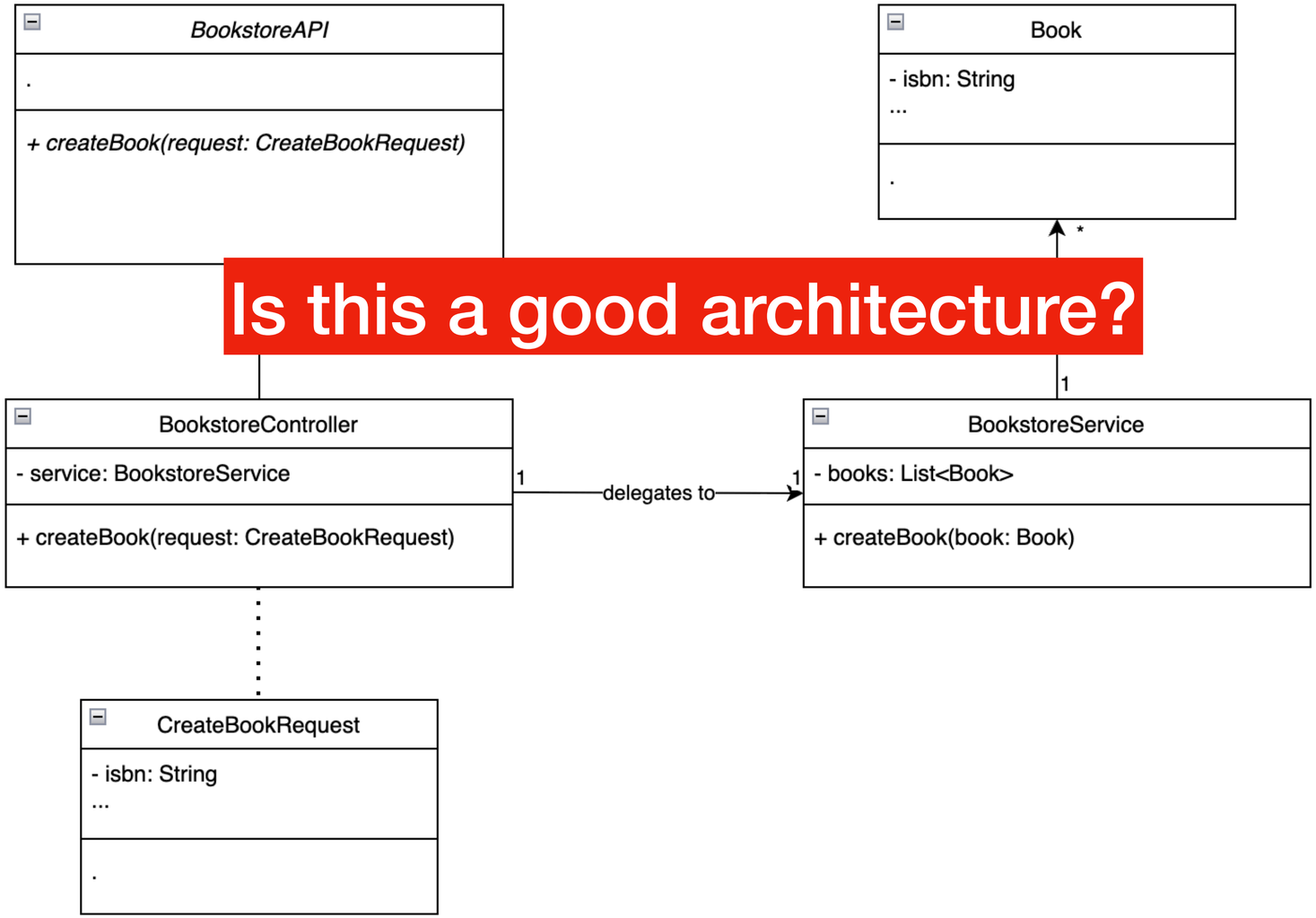
Software architecture



Is this a good architecture?

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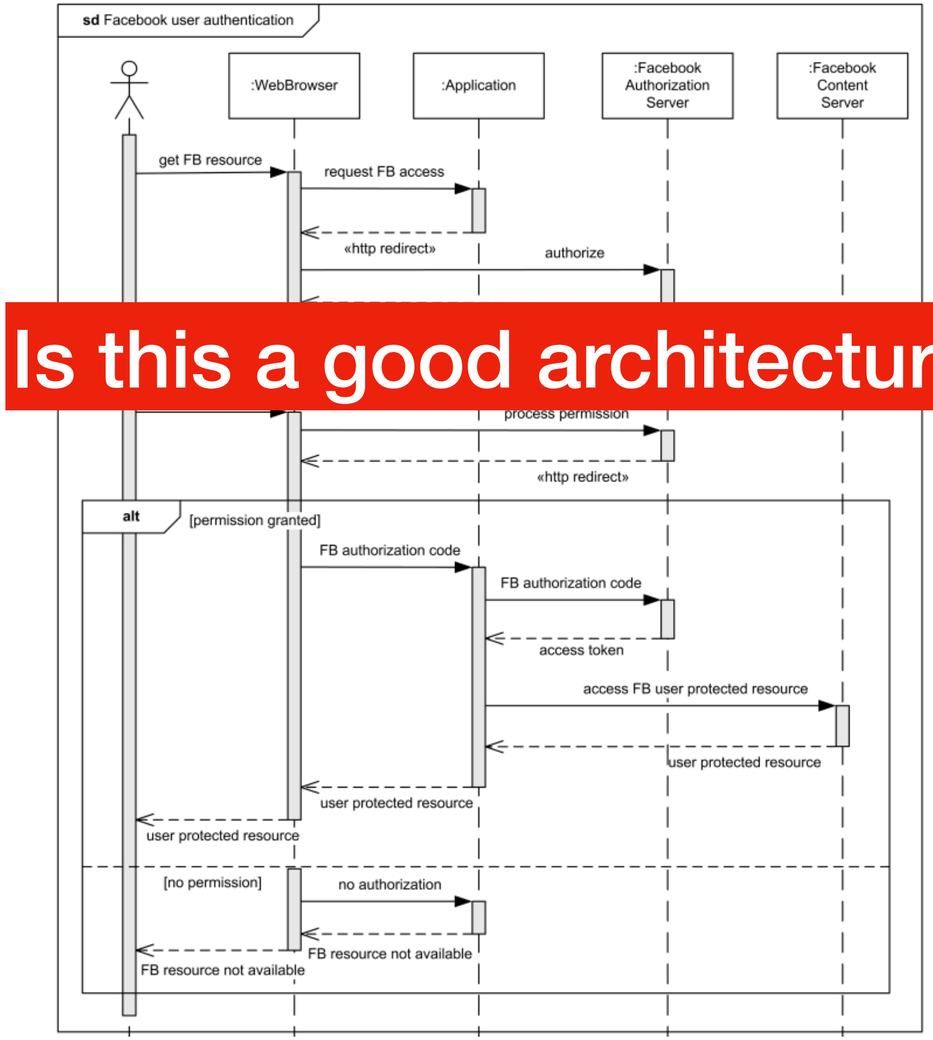
Software architecture



Is this a good architecture?

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Software architecture

What is a good architecture?

- Productive development
- Easy evolution/maintenance
- Trustworthiness (correct, secure, …)
- Performance

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Software architecture

- Implementation
  - module responsibilities, dependencies, type of relation (e.g., inheritance vs composition)
- Runtime
  - modules interactions, shared data stores, data flow, security requirements, performance
- Deployment
  - hardware/cloud infrastructure
- Organizational
  - module ownership

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Software architecture

- Implementation productive development
  - module responsibilities, dependencies, type of relation (e.g., inheritance vs composition)
- Runtime
  - modules interactions, shared data stores, data flow, security requirements, performance
- Deployment
  - hardware/cloud infrastructure
- Organizational productive development
  - module ownership

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Software architecture

- Implementation productive development easy evolution
  - module responsibilities, dependencies, type of relation (e.g., inheritance vs composition)
- Runtime
  - modules interactions, shared data stores, data flow, security requirements, performance
- Deployment easy maintenance
  - hardware/cloud infrastructure
- Organizational productive development
  - module ownership

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Software architecture

- Implementation productive development easy evolution
  - module responsibilities, dependencies, type of relation (e.g., inheritance vs composition)
- Runtime secure
  - modules interactions, shared data stores, data flow, security requirements, performance
- Deployment easy maintenance secure
  - hardware/cloud infrastructure
- Organizational productive development
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Software architecture

- Implementation productive development easy evolution
  - module responsibilities, dependencies, type of relation (e.g., inheritance vs composition)
- Runtime secure performance
  - modules interactions, shared data stores, data flow, security requirements, performance
- Deployment easy maintenance secure performance
  - hardware/cloud infrastructure
- Organizational productive development
  - module ownership

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Software architecture

Implementation

Define a system as a set of modules and their dependencies

module

A logical unit of software that groups related components behind a defined abstraction boundary

dependency

A relationship where one module relies on another to perform its responsibilities

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Modularization & Dependencies

```
fun main() {
    println("Hello World!")
}
```

- Single unit
- No dependencies
- No modularization needed
- Everything in one place

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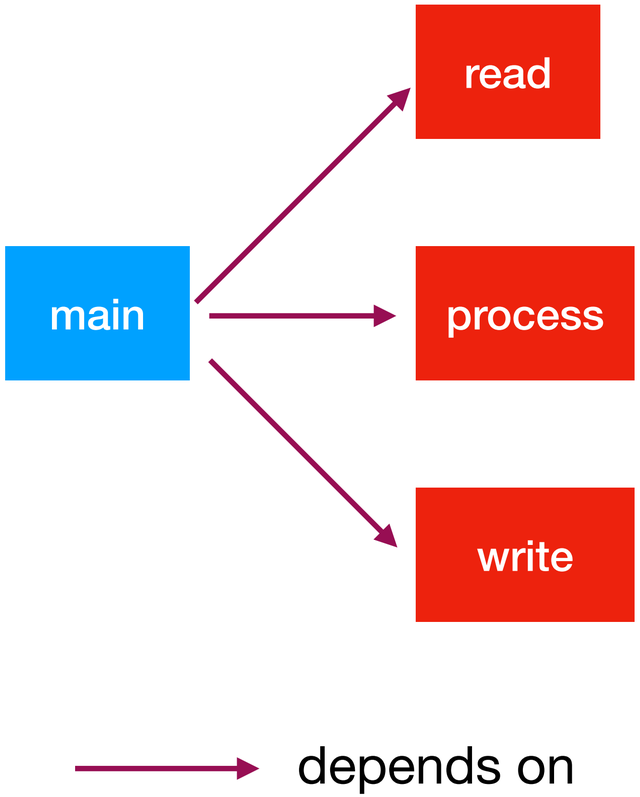
Modularization & Dependencies

```
fun main() {
    val content = read()
    val result = process(content)
    write(result)
}
```



- main <u>depends</u> on read, process and write
- benefits: reuse, testability, readability

functions = abstraction boundaries

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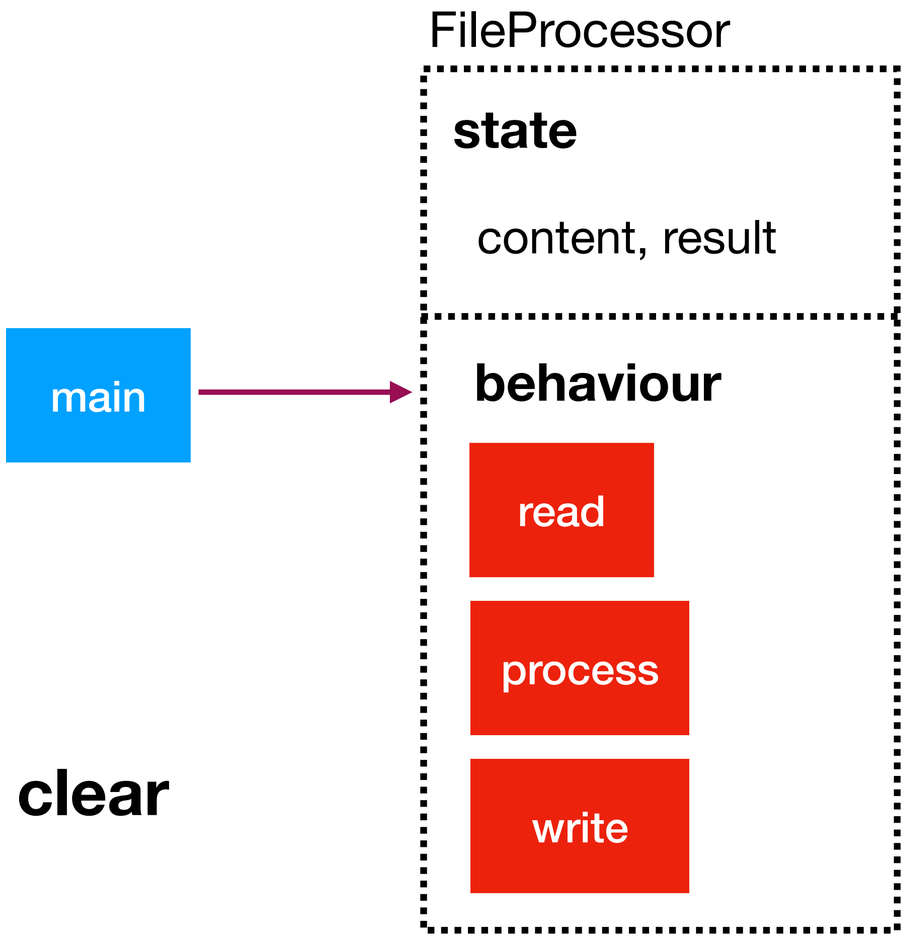
Modularization & Dependencies

```
fun main() {
    val fileProcessor = FileProcessor()
    fileProcessor.read()
    fileProcessor.process()
    fileProcessor.write()
}
```



- main <u>depends</u> on FileProcessor
- benefits: increased reuse, testability, readability, **clear ownership of state, separation of concerns**

classes = abstraction boundaries

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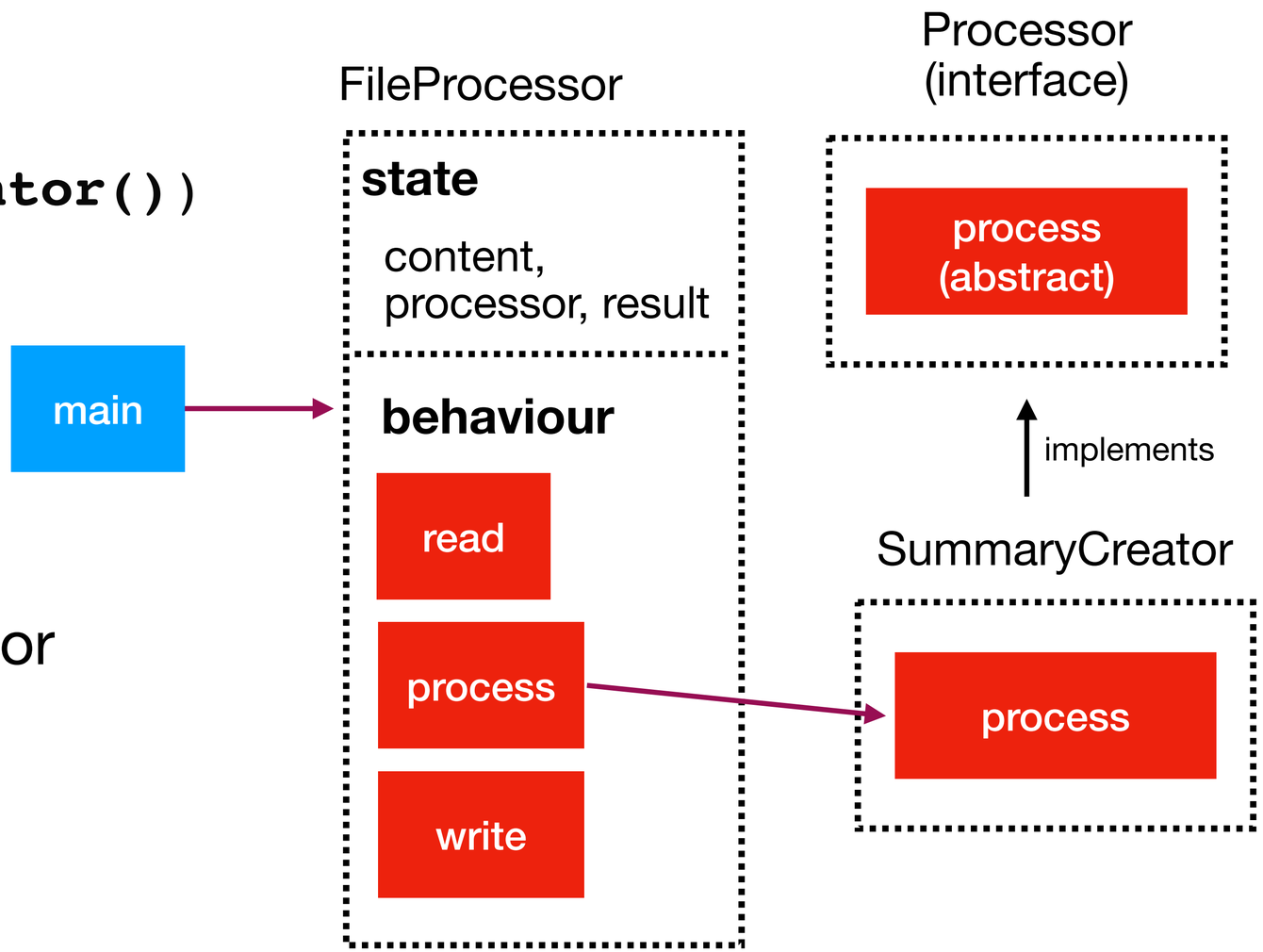
Modularization & Dependencies

```
fun main() {
    val fileProcessor =
      FileProcessor(SummaryCreator())
    fileProcessor.read()
    fileProcessor.process()
    fileProcessor.write()
}
```



- main <u>depends</u> on FileProcessor which depends on SummaryCreator (**dependency injection**)
- even more reusability!

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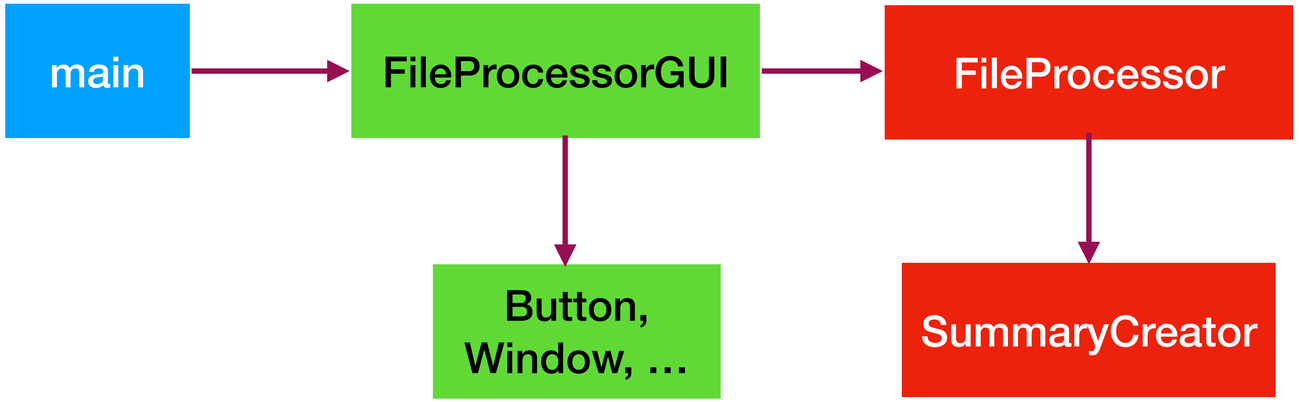
Modularization & Dependencies

```
fun main() {
    val fileProcessorGUI =
        FileProcessorGUI(
            FileProcessor(
                SummaryCreator()
            )
        )
    fileProcessorGUI.show()
}
```



- main depends on FileProcessorGUI which depends on FileProcessor

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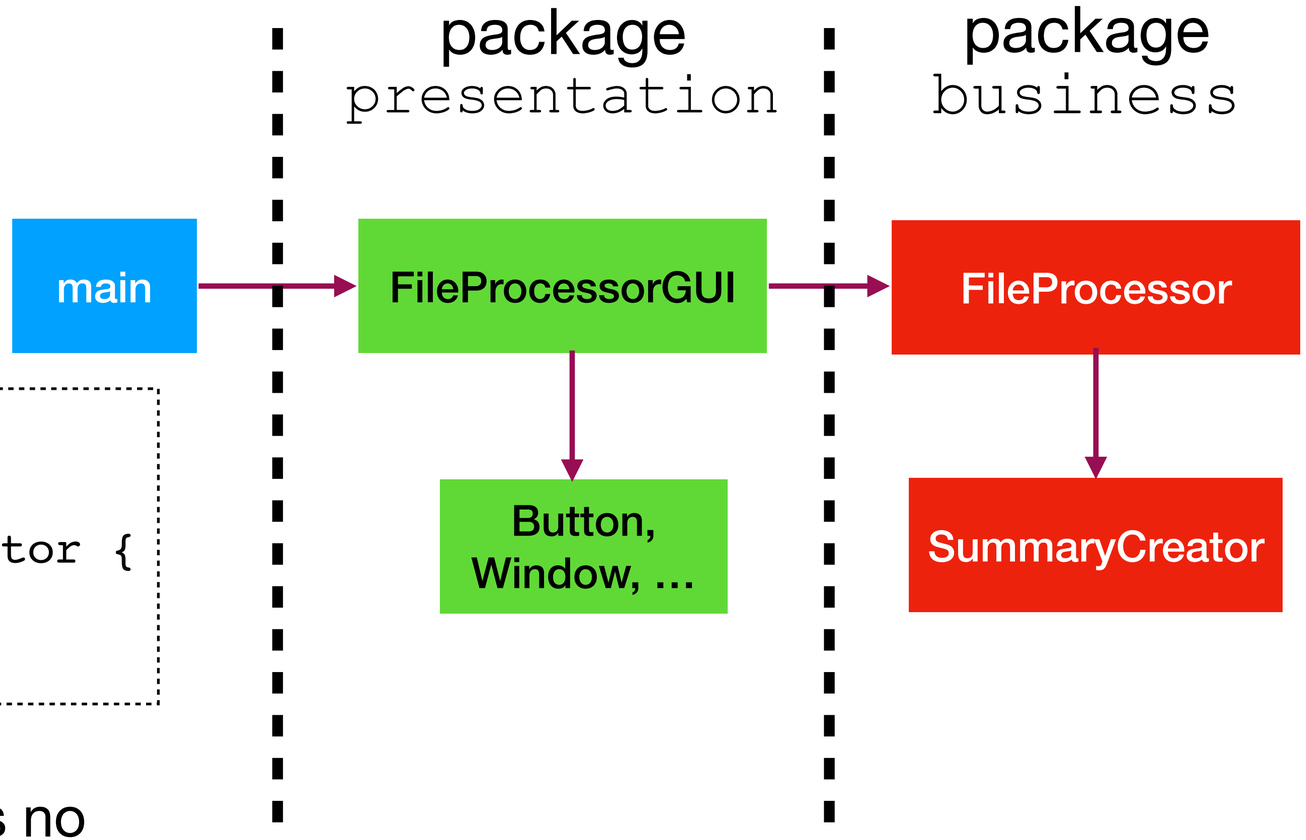
Modularization & Dependencies

```
package presentation

class FileProcessorGUI {
    ...
}
```

```
package business

class FileProcessor {
    ...
}
```

```
package business

class SummaryCreator {
    ...
}
```



packages are arbitrary sets of classes - there is no boundary enforcement (unless you use Spring Modulith)

packages = abstraction boundaries

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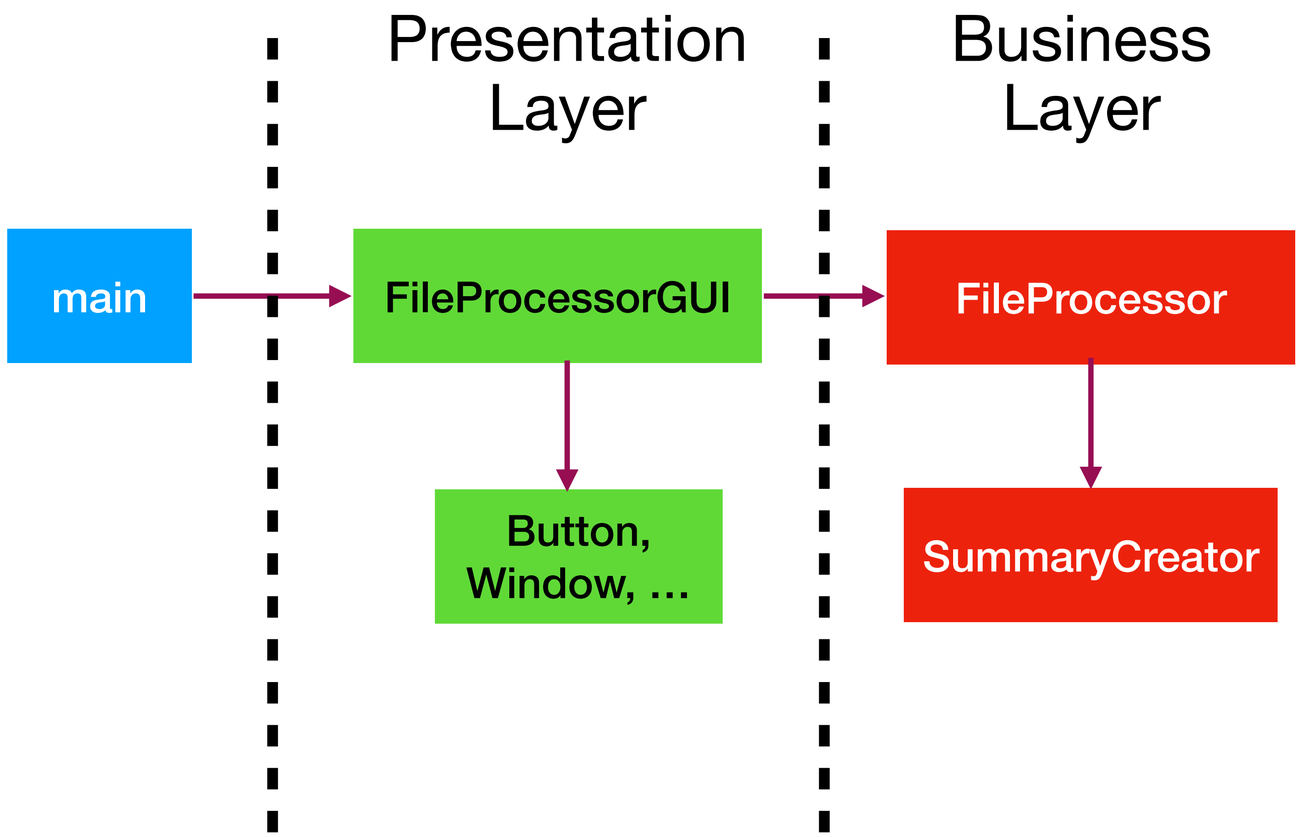
Modularization & Dependencies

```
fun main() {
    val fileProcessorGUI =
        FileProcessorGUI(
            FileProcessor(
                SummaryCreator()
            )
        )
    fileProcessorGUI.show()
}
```



- main depends on FileProcessorGUI which depends on FileProcessor

layers = abstraction boundaries

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Detour - Layers

Layered pattern

- Relation between layers is unidirectional
- Layers interact according to a strict ordering relation
- Layers interact only with the next layer (can’t jump layers)

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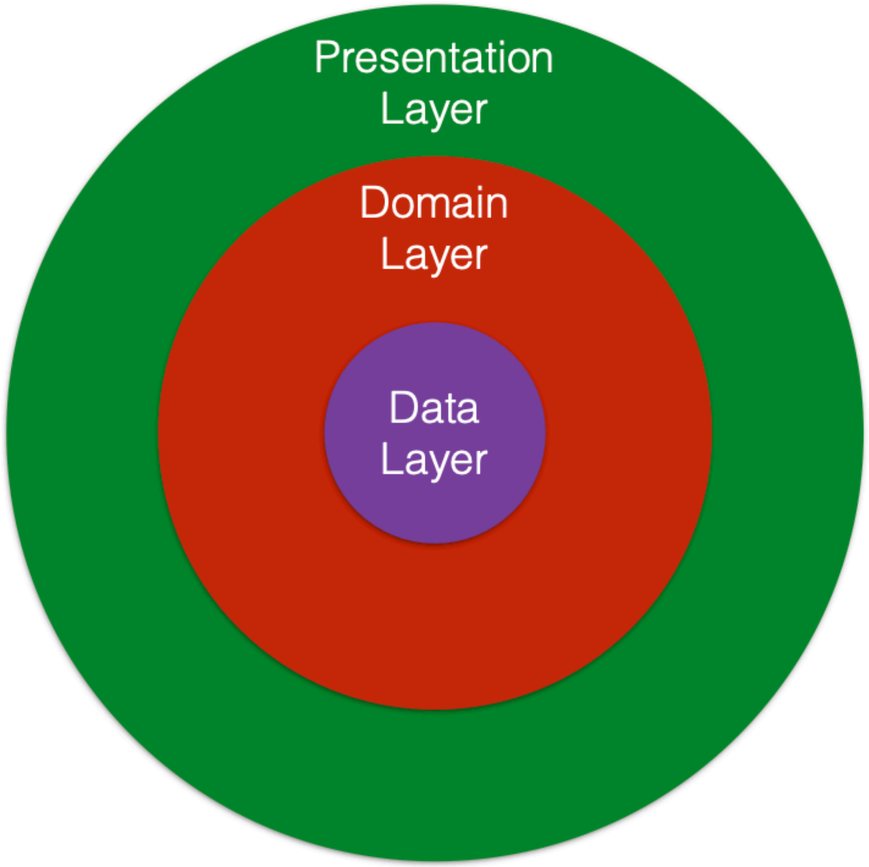
Detour - Layers

Data-centric applications



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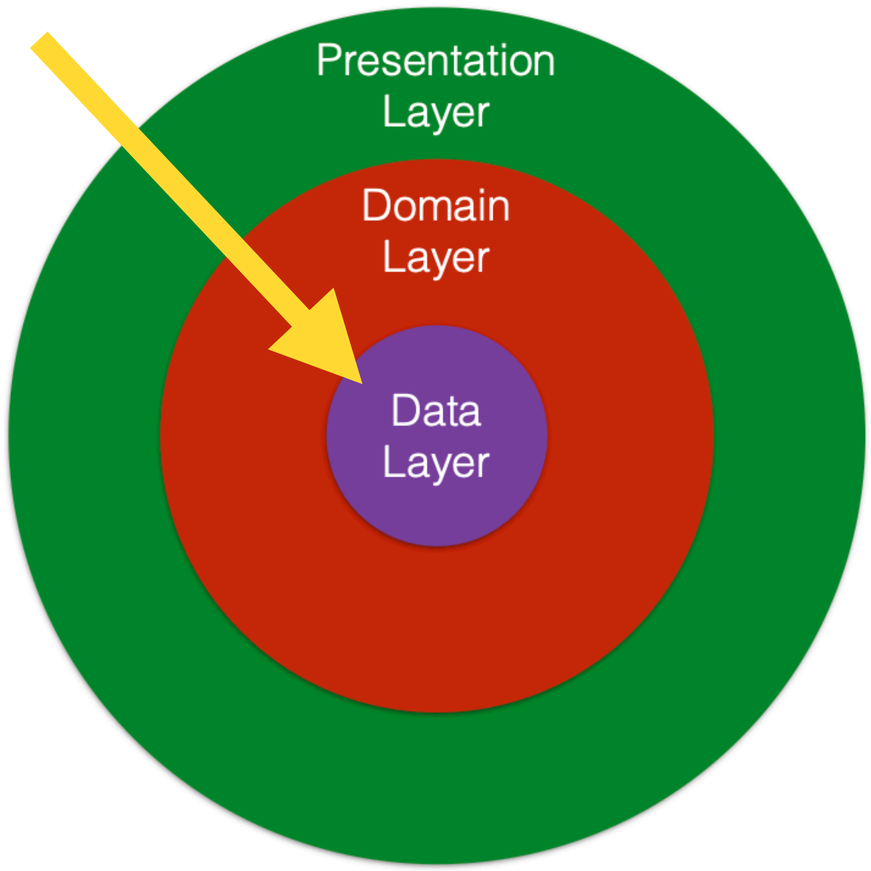
Detour - Layers

Interaction starts with the outer layer (Presentation Layer)

Presentation Layer must talk with the Domain Layer, can’t talk directly with Data Layer

Domain Layer must talk with the Data Layer



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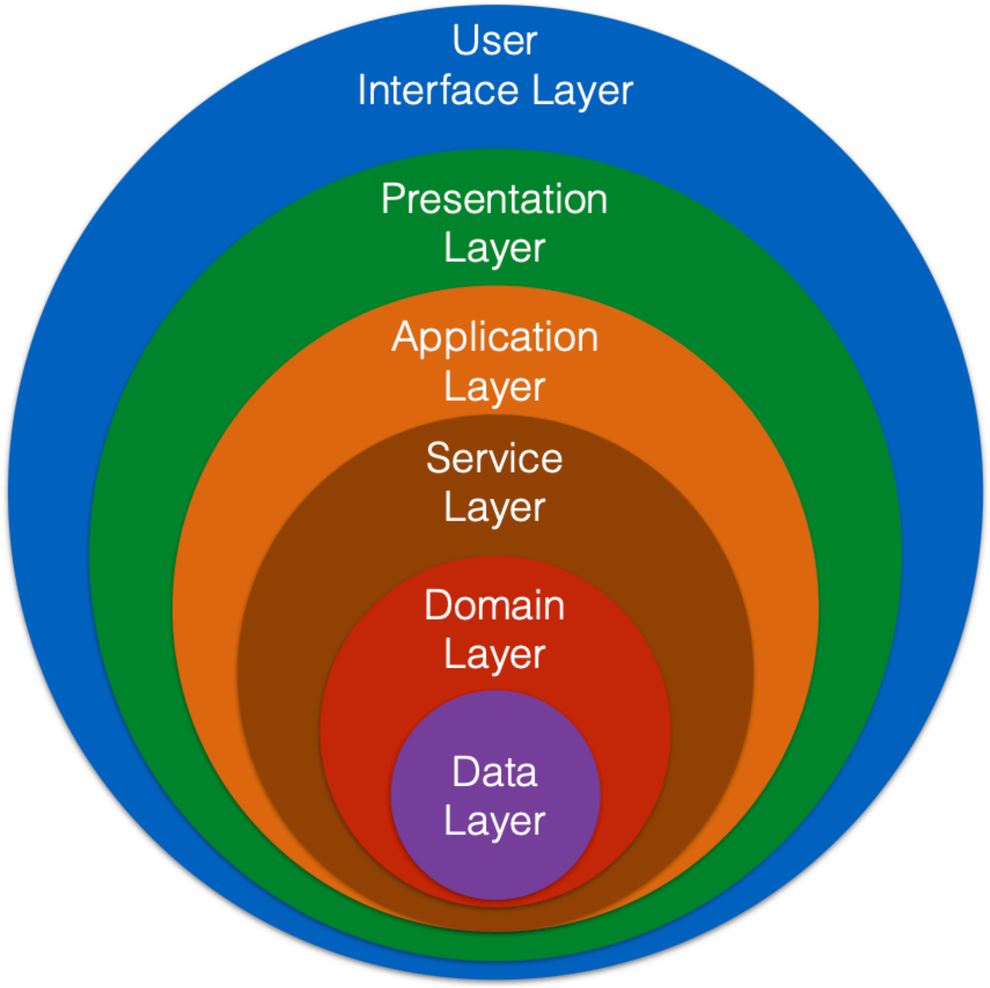
Detour - Layers



https://dzone.com/articles/layered-architecture-is-good

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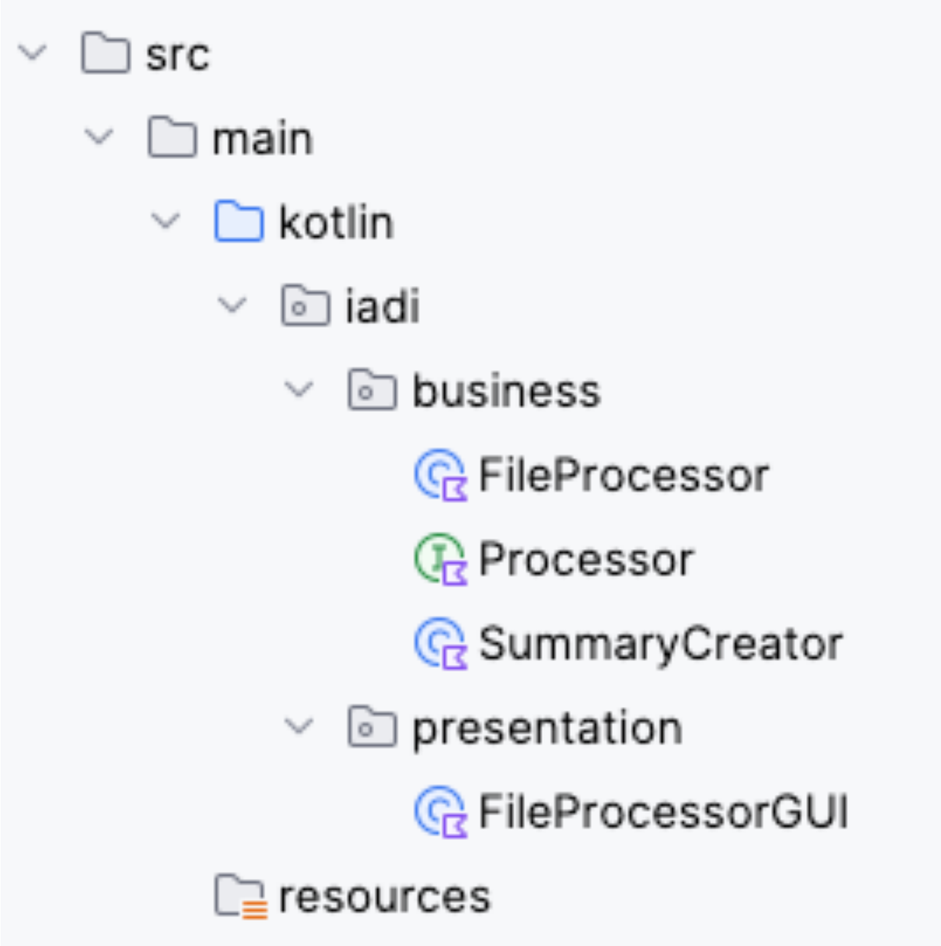
Modularization & Dependencies



Single code base

- benefits: atomic changes (refactoring, etc…), no versioning problems, simpler debugging, faster development
- problems: weak boundaries, difficult to scale across teams, no incremental deployment, long build times, increased merge conflicts

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Detour - Maven multi-module

```
my-project/
├── pom.xml (Parent / Aggregator POM)
│
├── common/
│   ├── pom.xml
│   └── src/
│       ├── main/
│       │   ├── java/
│       │   │   └── com/example/common/
│       │   └── resources/
│
├── service/
│   ├── pom.xml
│   └── src/
│       ├── main/
│       │   ├── java/
│       │   │   └── com/example/service/
│       │   └── resources/
│
└── web/
    ├── pom.xml
    └── src/
        ├── main/
        │   ├── java/
        │   │   └── com/example/web/
        │   ├── resources/
        │   └── webapp/
```

Maven Project with:

- Parent
  - Usually has only a POM
  - Defines shared dependencies
  - Lists modules
  - Builds everything together
- Subprojects (modules)
  - Each one has its own POM
  - References the parent
  - Defines specific dependencies
  - Can depend on other modules

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Detour - Maven multi-module

```
my-project/
├── pom.xml (Parent / Aggregator POM)
│
├── common/
│   ├── pom.xml
│   └── src/
│       ├── main/
│       │   ├── java/
│       │   │   └── com/example/common/
│       │   └── resources/
│
├── service/
│   ├── pom.xml
│   └── src/
│       ├── main/
│       │   ├── java/
│       │   │   └── com/example/service/
│       │   └── resources/
│
└── web/
    ├── pom.xml
    └── src/
        ├── main/
        │   ├── java/
        │   │   └── com/example/web/
        │   ├── resources/
        │   └── webapp/
```

sub-module

sub-module

sub-module

maven sub-modules = abstraction boundaries

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Detour - Maven multi-module

/pom.xml

```
<project>
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>
  <artifactId>my-project</artifactId>
  <version>1.0.0</version>
  <packaging>pom</packaging>

  <modules>
    <module>common</module>
    <module>service</module>
    <module>web</module>
  </modules>

  <dependencyManagement>
    <dependencies>
      <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-core</artifactId>
        <version>6.1.0</version>
      </dependency>
    </dependencies>
  </dependencyManagement>
</project>
```

list of modules

shared dependencies

/service/pom.xml

```
<project>
  <parent>
    <groupId>com.example</groupId>
    <artifactId>my-project</artifactId>
    <version>1.0.0</version>
  </parent>

  <artifactId>service</artifactId>

  <dependencies>
    <dependency>
      <groupId>com.example</groupId>
      <artifactId>common</artifactId>
      <version>${project.version}</version>
    </dependency>
  </dependencies>
</project>
```

reference to parent

depends on module “common”

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Detour - Maven multi-module

In the project base folder:

```
mvn compile
```

Compiles all modules, manages dependencies between modules

```
mvn compile -pl service
```

Compiles just the service module

```
mvn compile -pl service -am
```

Compiles the service module and all its module dependencies

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Detour - Maven multi-module

maven single-module

```
file-processor/
├── pom.xml
└── src/
    └── main/
        └── kotlin/
            └── iadi/
                ├── business/
                │   ├── FileProcessor.kt
                │   ├── Processor.kt
                │   └── SummaryCreator.kt
                └── presentation/
                    └── FileProcessorGUI.kt
```

maven multi-module

```
file-processor-parent/
├── pom.xml(parent / aggregator)
│
├── business/
│   ├── pom.xml
│   └── src/
│       ├── main/
│       │   └── kotlin/
│       │       └── iadi/
│       │           └── business/
│       │               ├── FileProcessor.kt
│       │               ├── Processor.kt
│       │               └── SummaryCreator.kt
│
└── presentation/
    ├── pom.xml
    └── src/
        ├── main/
        │   └── kotlin/
        │       └── iadi/
        │           └── presentation/
        │               └── FileProcessorGUI.kt
```

It's still a single code base

- benefits: atomic changes (refactoring, etc…), no versioning problems, simpler debugging, faster development
- **less** problems: **not so** weak boundaries, **not so** difficult to scale across teams, **allows** ~~no~~ incremental deployment, long build times, **slightly less** merge conflicts

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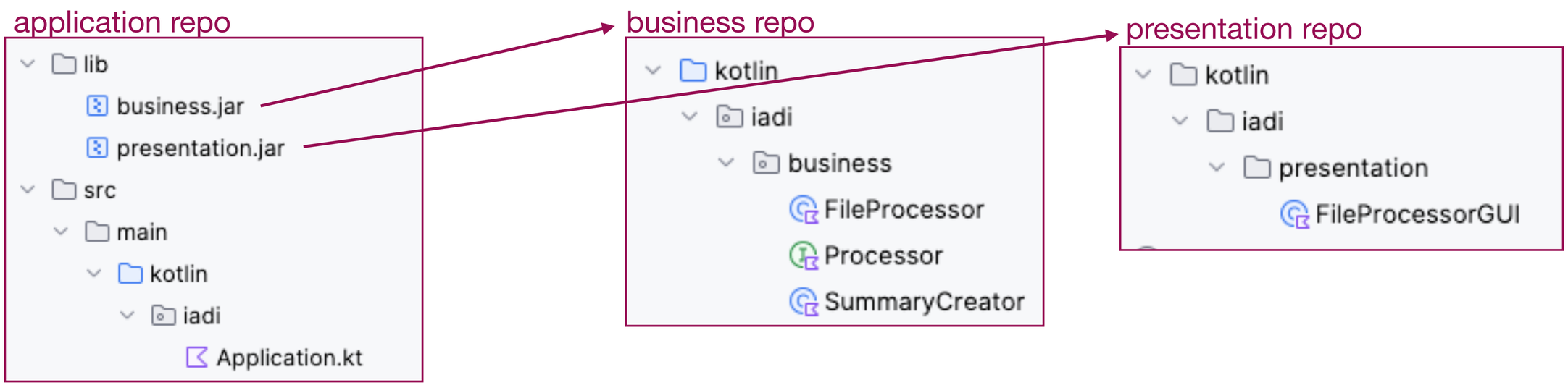
Modularization & Dependencies



Different code bases => Different libraries (developed in-house)

- benefits: stronger module boundaries, independent teams, independent releases, clear ownership
- problems: versioning complexity, slower refactorings, dependency hell, over-modularization risks

libraries = abstraction boundaries

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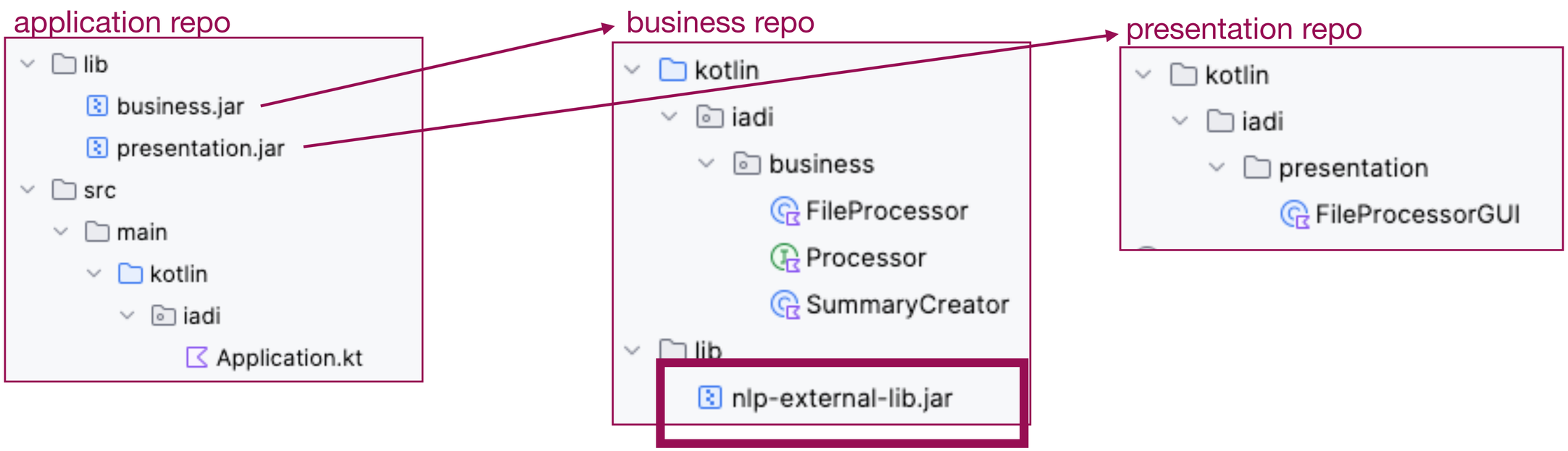
Modularization & Dependencies



Libraries (externally developed)

- benefits: “free” productivity, higher quality (often), standardization
- problems: dependency hell x 10, hidden complexity, harder debugging, lock-in

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Detour - dependency management

Usually, dependencies are not managed manually (i.e., by putting JAR files inside a lib folder) but rather managed by a dependency management tool such as Maven, Gradle, pip, or npm.

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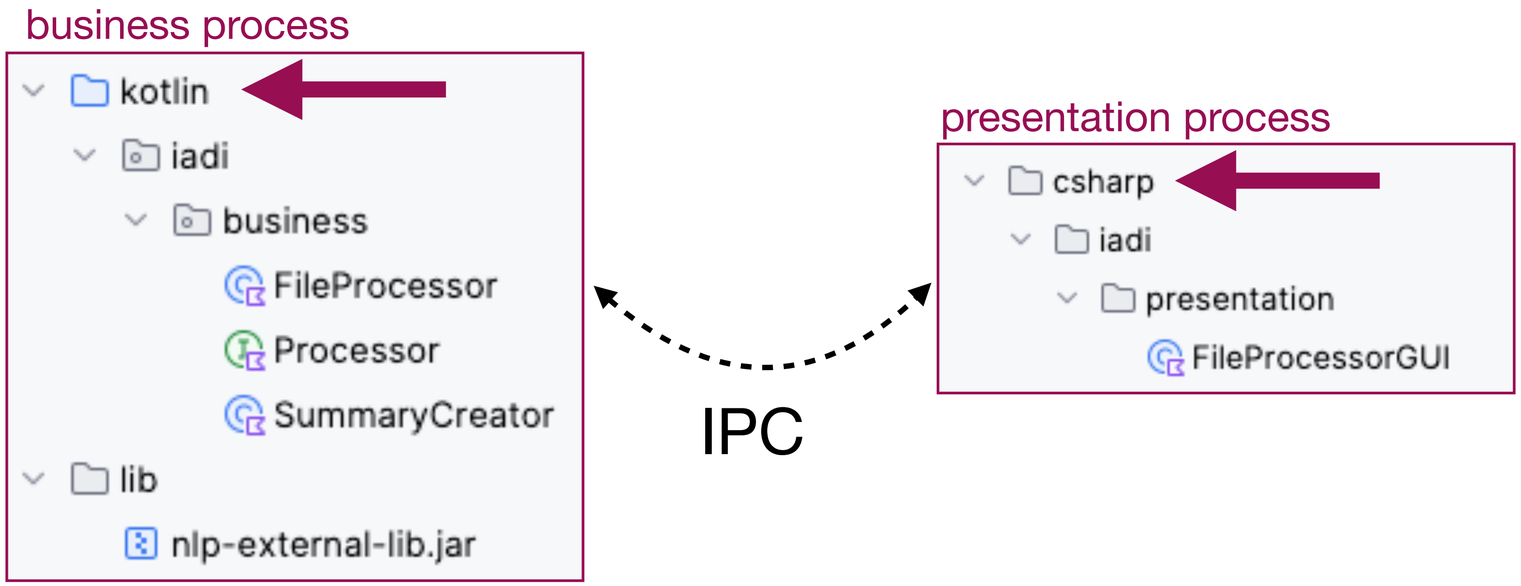
Modularization & Dependencies



Separate OS processes (same machine)

Communication through IPC sockets, shared memory, common database

- benefits: language/runtime independence, fault isolation
- problems: serialization, deployment complexity

stronger decoupling through process boundaries

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Modularization & Dependencies

Here comes Internet….

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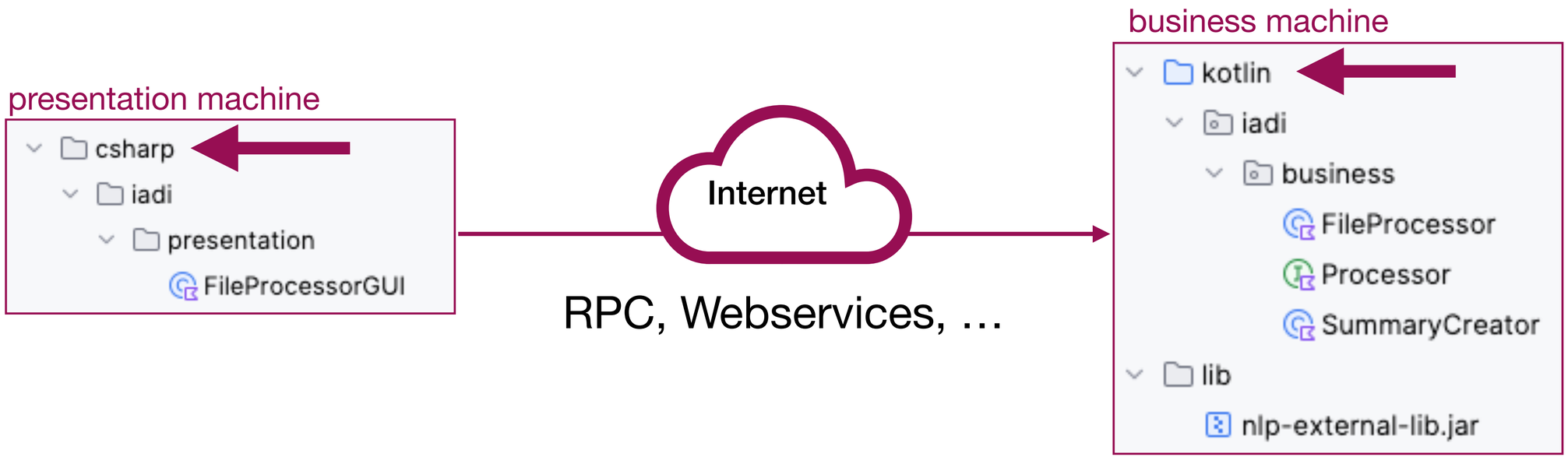
Modularization & Dependencies



Client-server using native

- benefits: independent deployment, centralization of logic, easier updates (server-side), multiple clients supported
- problems: latency, network failures, security, not scalable to multiple domains

stronger decoupling through network boundaries

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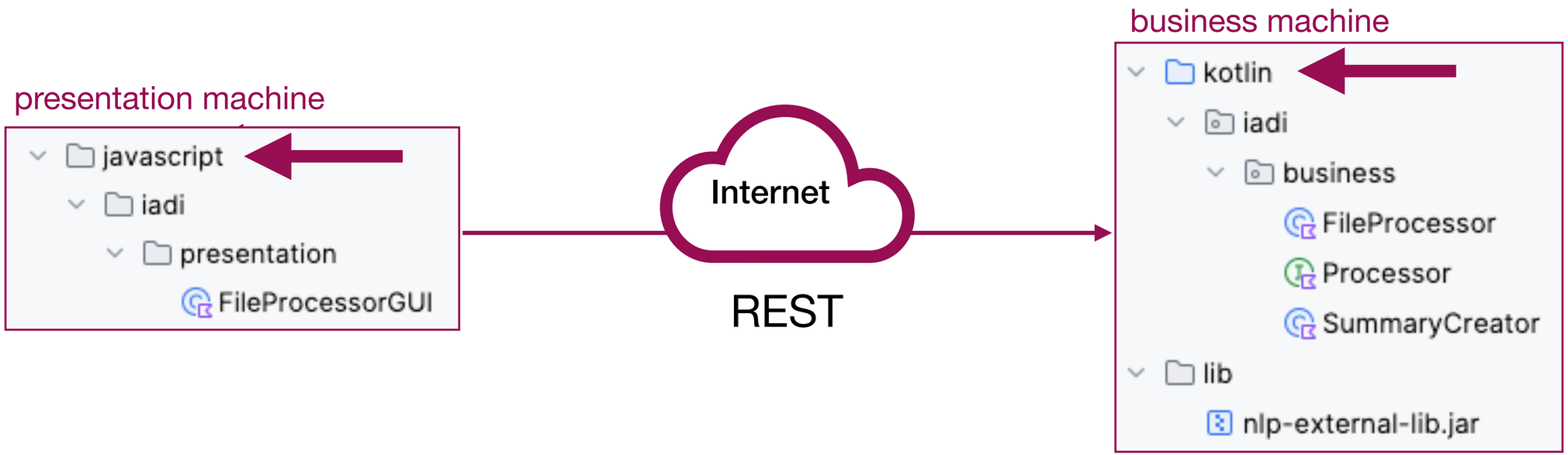
Modularization & Dependencies



Client-server using the browser

- benefits: no need to install in the client, always up-to-date
- problems: language must be javascript(*), restricted environment, not a native UI

(*) Or a language that transpiles to javascript such as typescript

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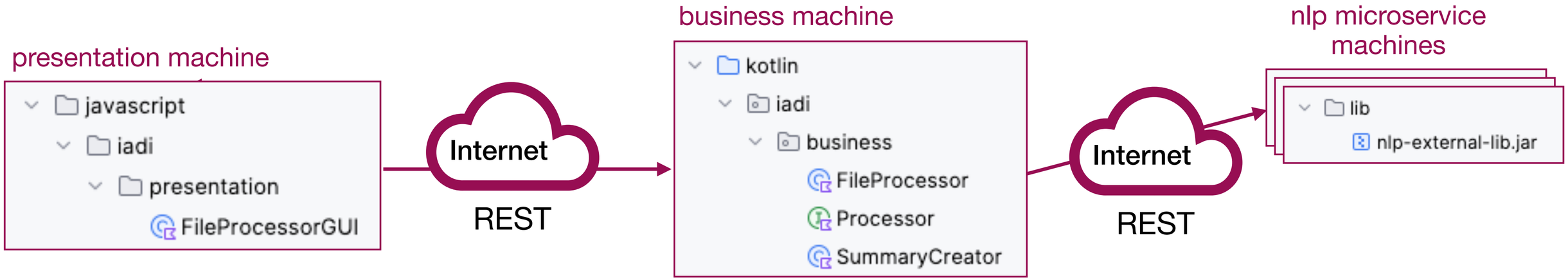
Modularization & Dependencies



Micro-services

- benefits: domain boundaries, team decentralization, deployment autonomy, horizontal scaling
- problems: the same of client-server + complexity

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Detour - Scaling

Vertical scaling

- **Make one machine stronger (increase CPU, memory, …)**
- Benefits: simplest approach, no architectural changes, easier debugging, no replication logic
- Problems: hardware limits, expensive, single point of failure
- Mostly used in Client-Server

Horizontal scaling

- **Add more machines**
- Benefits: theoretically unlimited scaling, cheaper hardware, fault tolerance
- Problems: complexity, consistency and synchronization problems, operational overhead
- Mostly used in micro-services

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Modularization & Dependencies

In summary

- Functions
- Classes
- Packages
- Layers
- Libraries (*)
- Separate processes
- Client-Server
- Micro-services

small modules

- ✅ Isolation
- ✅ Team distribution
- ✅ Deployment independence

- 👎 Increased complexity
- 👎 Network failures
- 👎 Higher need for team communication

large modules

(*) Includes maven multi-module

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Exercise

	Benefit over previous
everything in main()	-
Functions	readability, reuse
Classes	
Packages	
Layers	
Libraries	
Separate processes	
Client-server	
Micro-services	

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Resolution

	Benefit over previous
everything in main()	-
Functions	readability, reuse
Classes	encapsulate state + behavior
Packages	organize related classes
Layers	enforce dependency rules: unidirectional, ordered
Libraries	independent code-bases, team autonomy
Separate processes	different runtime/language, fault isolation
Client-server	independent deployment between client and server
Micro-services	independent deployment between server modules

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